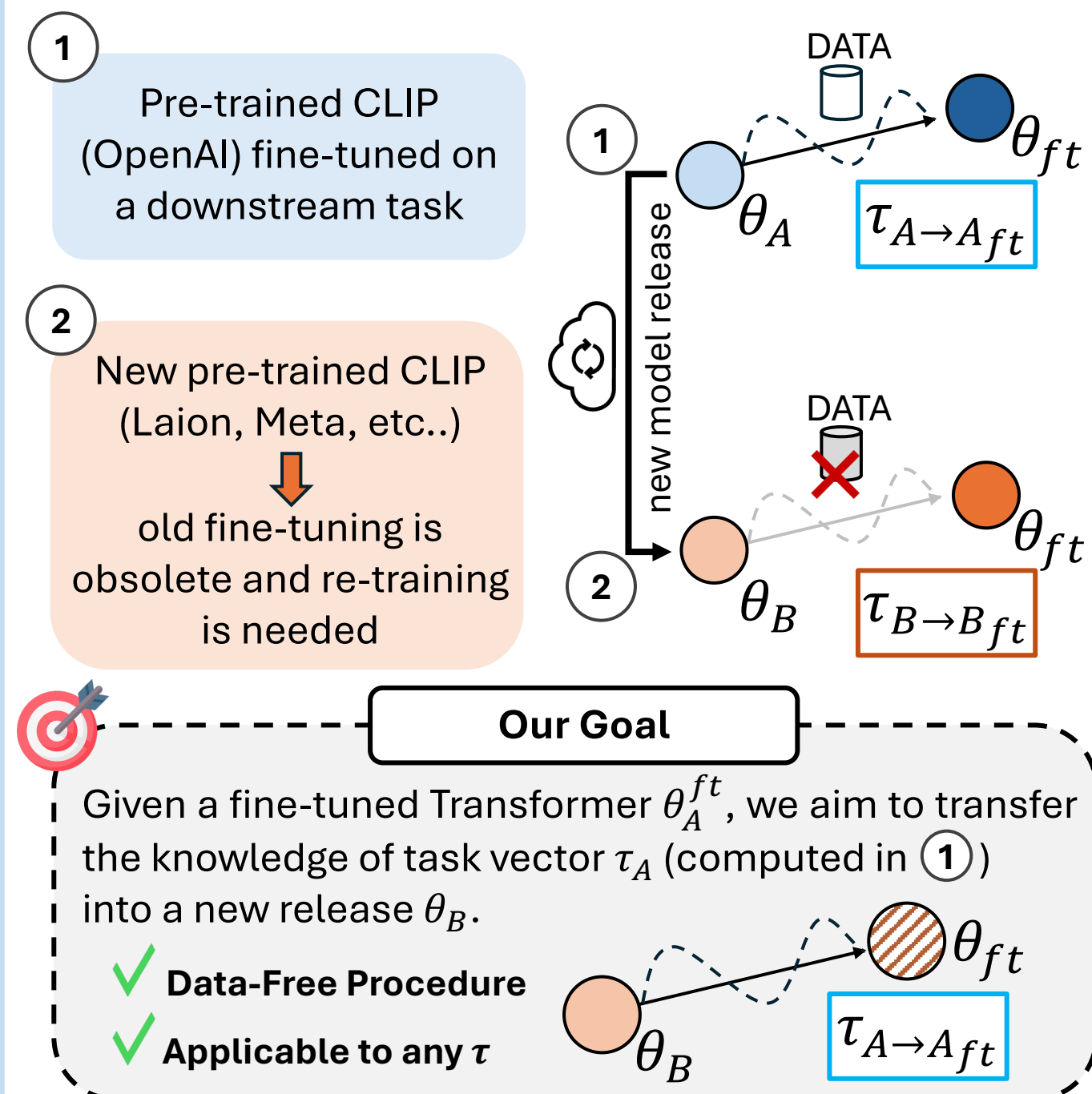


Update Your Transformer to the Latest Release: Re-Basin of Task Vectors

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MOTIVATION



BACKGROUND

Model Re-Basin: Exploits permutation symmetry to align different trained models into a shared optimization basin, enabling their interpolation. In our case, we need to align θ_A with θ_B to mount τ_A on θ_B .

Functional Equivalence: NNs exhibit permutation symmetry due to the exchangeability of units within layers. For an MLP layer with activation σ , applying permutation matrix P yields:

$$z_{\ell+1} = \sigma(W_{\ell}z_{\ell} + b_{\ell}) = z_{\ell+1} = P^{\top} \sigma(PW_{\ell}z_{\ell} + Pb_{\ell})$$

Thus, preserving functional equivalence requires applying consistent permutations across the network:

$$W'_{\ell} = PW_{\ell}, \quad b'_{\ell} = Pb_{\ell}, \quad W'_{\ell+1} = W_{\ell+1}P^{\top}$$

⚠ **Limitations:** Fail with multi-head attention structure.

RE-BASIN OF TASK VECTORS

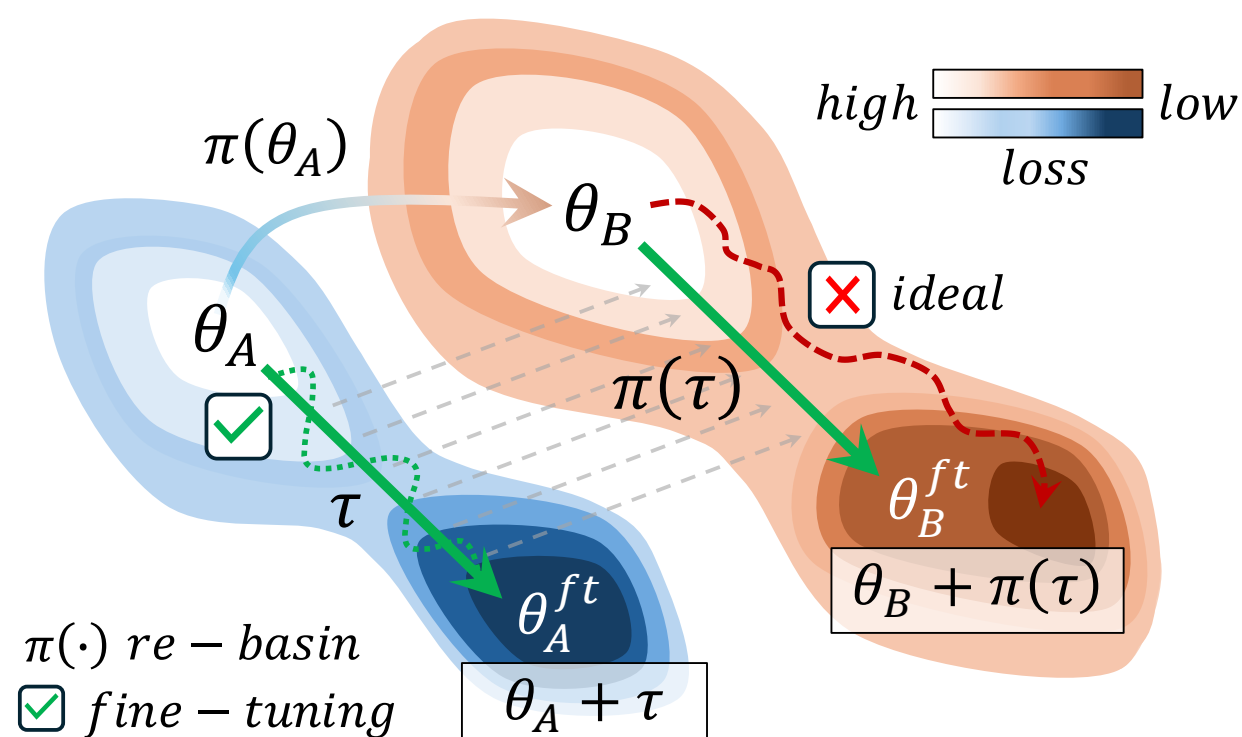
TransFusion permutes Transformer weights while preserving attention and residual structures, enabling transfer of task-specific updates across different pretrained model versions.

Hybrid Weight Permutation Approach

- **MHA layers:** Custom spectral alignment
- **Other layers:** Git Re-Basin alignment

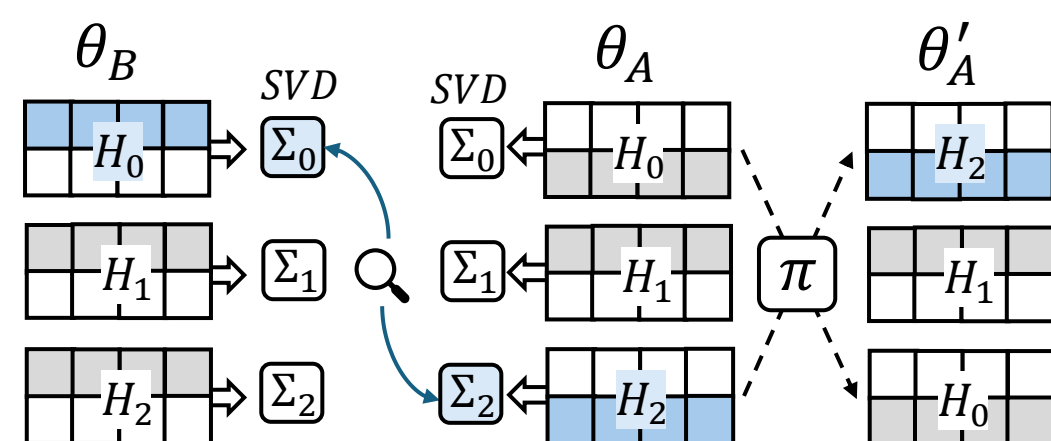
Task-Vector Transport

After computing best permutations π , such that $\pi(\theta_A) \approx \theta_B$, we transport the task vector on the new backbone θ_B :

$$\theta_B^{ft} = \theta_B + \pi(\tau_A)$$


WEIGHTS MATCHING

A two-level permutation strategy that first finds optimal mappings between pairs of heads (**Inter-Head matching**), then refines permutations within those matched heads (**Intra-Head matching**).



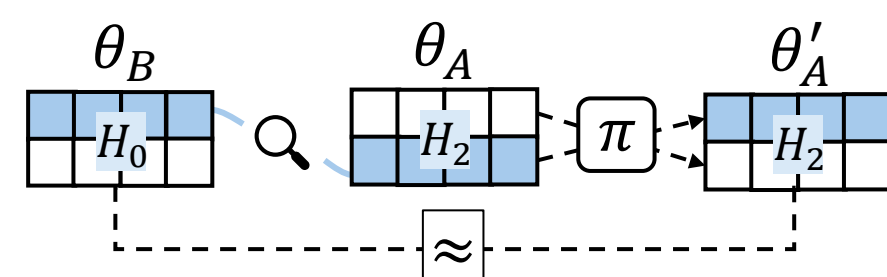
Inter-Head Matching: Find optimal pairing π between attention heads across models using a spectral metric based on singular values.

$$\text{SVD}(W_h) = U_h \Sigma_h V_h^{\top}$$

$$d(h_i^{(A)}, h_j^{(B)}) = \|\Sigma_i^{(A)} - \Sigma_j^{(B)}\|_F$$

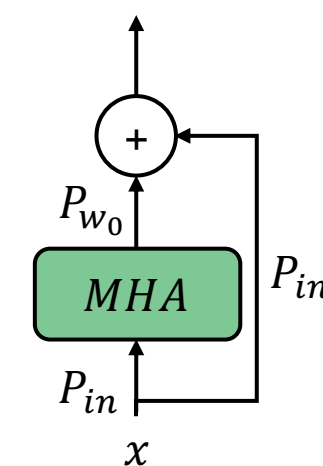
Proposition Let $h \in \mathbb{R}^{m \times n}$ and define $d_p(h_1, h_2) = \|\sigma(h_1) - \sigma(h_2)\|_p$, where $\sigma(h)$ is the vector of singular values. For permutation matrices P_r, P_c , we have:

$$\sigma(P_r h P_c) = \sigma(h) \Rightarrow d_p(h, P_r h P_c) = 0.$$



Intra-Head Matching: Determine permutations π that maximize the inner products across projection weight partitions corresponding to each matched head pair.

FUNCTIONAL EQUIVALENCE



Residual Connection

Ensure consistency by aligning identity paths with attention permutations:

$$z_i = P_{W_0} z_{attn} + I_i P_{in} x = P_{W_0} z_{attn} + P_{W_0} x$$

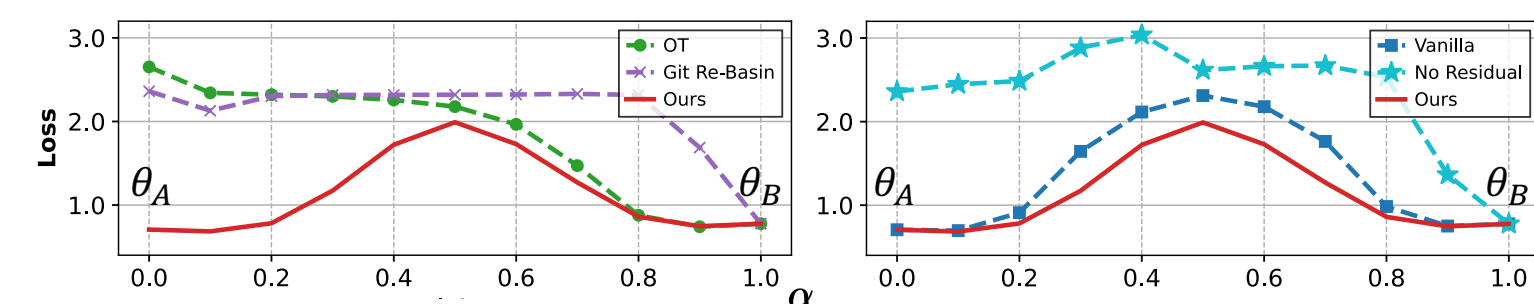
Multi-Head Attention

The output of MHA is **equivariant** to TransFusion permutations:

- Permutation Inter-Head: $P_{inter_head} \in \pi$
- Permutation Intra-Head: $P_{intra_head} \in \pi, P_{intra_head} = \{P^{(i)}\}_{i=1}^H$
- MHA Block Permutation: $P_{attn} = P_{inter_head} \circ P_{intra_head} \in \pi$

$$O' = MHA(X; P_{attn} W_q, P_{attn} W_k, P_{attn} W_v) = O P_{attn}$$

$$O = O' P_{attn}^{\top}$$



EXPERIMENTS

Method	EUROSAT		DTD		GTSRB		SVHN	
	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.
θ_B zero-shot	49.02	68.73	47.50	68.73	43.42	68.73	45.97	68.73
$\theta_B + \tau$	-7.62	-16.15	-0.15	-0.10	-5.39	-0.70	-22.00	-16.45
$\theta_B + \pi(\tau)$ (Optimal Transport)	-14.05	-5.28	-0.53	-1.18	-2.43	-1.30	-12.30	-2.70
$\theta_B + \pi(\tau)$ (Git Re-Basin)	+0.95	-0.48	-0.91	-0.02	+0.76	-0.05	+0.79	+0.30
TRANSFUSION (OURS)	+4.95	-0.06	+0.21	-0.08	+1.10	-0.40	+3.64	-0.48

For all experiments we consider **CLIP ViT-B/16**. We use CommonPool pre-training for θ_A and Datacomp for θ_B . Our method boosts θ_B zero-shot performance and preserves generalization in the updated model.

Zero-shot gain/drop relative to θ_B of naive $\theta_B + \tau$ and our strategy $\theta_B + \pi(\tau)$ varying α .

Method	QQP				SST2				RTE				CoLA			
	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.	TASK	SUPP.
θ_B	55.00	50.69	54.51	40.94	-8.29	+0.23	-2.53	-0.77	-8.31	+5.39	-1.08	-1.25	-8.31	+5.39	-1.08	-1.25
$\theta_B + \tau$	-8.29	+0.23	-2.53	-0.77	-8.31	+5.39	-1.08	-1.25	-8.31	+5.39	-1.08	-1.25	-8.31	+5.39	-1.08	-1.25
$\theta_B + \pi(\tau)$ (OT)	+3.58	+5.73	+2.17	+1.44	+3.58	+5.73	+2.17	+1.44	+3.58	+5.73	+2.17	+1.44	+3.58	+5.73	+2.17	+1.44
TRANSFUSION (OURS)	+6.50	+5.96	+3.61	+2.49	+6.50	+5.96	+3.61	+2.49	+6.50	+5.96	+3.61	+2.49	+6.50	+5.96	+3.61	+2.49

